

Oleksii Krasnoshtanov

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EDUCATION

Breda University of Applied Sciences

BSc Applied Data Science & Artificial Intelligence; 9.1/10 (4.0 GPA); BUas Scholarship

Breda, Netherlands

Sep 2024 – Present

PROJECTS & TECHNICAL WORK

CV Pipeline – Plant Phenotyping Platform | *5-person group project, BUas* | [Code](#) | [Docs](#)

Apr – Jun 2026

- Built an MLOps platform for plant organ segmentation and root-tip detection: U-Net (ResNet-34) behind an authenticated FastAPI backend, with a Next.js frontend, PostgreSQL and Prometheus
- Ran the model lifecycle from Airflow DAGs: data versioning, Azure ML training, drift detection on confidence, latency and error rate, and retraining driven by researcher feedback on flagged predictions
- Owned deployment and CI/CD: one codebase shipping to Docker Compose, on-premise Portainer and Azure Container Apps, with blue-green cutover, health-gated rollback and OIDC cloud auth; 85% coverage gate, 430 tests

Dutch Sign Language Fingerspelling Trainer | *Personal Project* | [Code](#) | [Docs](#)

May – Sep 2026

- Real-time NGT alphabet trainer: MediaPipe landmarks feed a PyTorch residual MLP for the 24 static letters (95% validation) and a bidirectional LSTM over 30-frame sequences for the two needing motion (92%)
- Sole author: FastAPI WebSocket backend streaming to a Next.js 16 / TypeScript frontend, bilingual EN/NL, containerised with CI and published documentation

NPEC Plant Phenotyping System | *Individual Project, BUas* | [Code](#)

Nov 2025 – Jan 2026

- Ranked **#2 of 77** in Kaggle competition for plant root segmentation; selected to present results
- Built the full autonomous loop – U-Net root segmentation, root-tip extraction, pixel-to-robot transform and OT-2 pipette control in PyBullet – benchmarking PID against a PPO reinforcement-learning controller

Emotion Analysis Pipeline | *Group Project – Dutch Media Client*

Sep – Nov 2025

- Production NLP pipeline for a Dutch media client, processing 40-minute episodes end-to-end: Faster-Whisper V3 transcription with VAD, multilingual emotion classification and automated visualisation
- Fine-tuned DeBERTa and Russian transformer models on merged datasets augmented with Mistral-generated synthetic data; cross-validated across languages to stabilise multilingual predictions

MARBET Event Support Chatbot | *2-Person Sprint* | [Code](#)

Apr – May 2025

- Built self-hosted RAG system (Llama 3.3 70B via Ollama, LangChain, FAISS) for corporate event Q&A with multi-format ingestion and Gradio web interface; client deployed to production

HACKATHONS & COMPETITIONS

BrabantHack 2026 | *Winner – Demcon Deep Tech Track* | [Code](#)

Apr 2026

- Won the Demcon track with a model that locates a pedestrian standing entirely outside the camera frame from the shadow they cast into it: **0.626 mean IoU** on the organisers' hidden test set; team of 3
- Drove the gain by re-describing the box relative to the frame edge it hides behind – side, distance, width, height, y-centre – on a ResNet-50 trunk fused with 19 shadow descriptors: 0.468 to 0.610 mean IoU, reproducible from the repository

HERE Technologies Hackathon | *RDM Verification Challenge*

2026

- Built XGBoost classification pipeline for Restricted Driving Manoeuvre verification combining probe GPS features with Mapillary and OSM geospatial signals; designed synthetic data scrambling for NDA-compliant delivery

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, HuggingFace Transformers

Computer Vision: OpenCV, U-Net, MediaPipe, PyBullet

NLP & Audio: LangChain, Faster-Whisper, FAISS, NLTK, Ollama

MLOps & Cloud: Azure ML, Azure Container Apps, Airflow, MLflow, Prometheus, GitHub Actions CI/CD

Infrastructure: Docker, Docker Compose, Portainer, FastAPI, Next.js, PostgreSQL, Git

Data & Analytics: Pandas, NumPy, Power BI

Spoken Languages: English (fluent), Ukrainian (native), Russian (fluent)